

INTERNSHIP REPORT

B.Tech in Computer Science and Engineering

DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING

CUSAT



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MAY 21 2025 - JUNE 30 2025



CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that **Afnash Ali P** has successfully completed the **Software Development Internship** program at **Cognifyr** from **21 May 2025** to **30 June 2025**. During this period, Afnash demonstrated exceptional dedication, enthusiasm, and a strong work ethic.

Throughout the internship, Afnash gained hands-on experience in designing, developing, and testing software applications, collaborating with cross-functional teams, and adhering to industry best practices.

We commend Afnash for their professionalism, teamwork, and commitment to delivering high-quality results. Their contributions to our projects were invaluable, and we are confident that they will excel in their future endeavours. This certificate is awarded in recognition of their outstanding performance and completion of the internship program.

Ajnas N B
Engineering Head
Cognifyr.CO

DATE: 25 July 2025

DIVISION OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

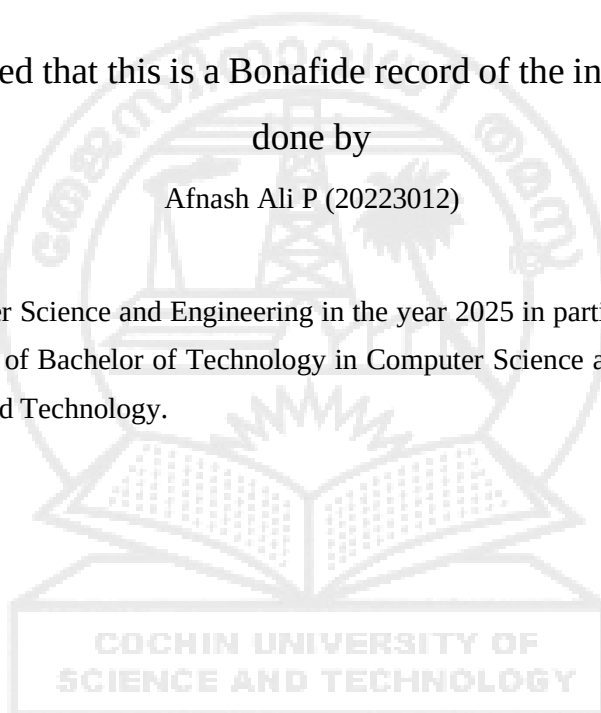
CERTIFICATE

Certified that this is a Bonafide record of the internship

done by

Afnash Ali P (20223012)

of V Semester, Computer Science and Engineering in the year 2025 in partial fulfilment requirements for the award of Degree of Bachelor of Technology in Computer Science and Engineering of Cochin University of Science and Technology.



Dr. Pramod Pavithran
Head Of Division

Dr. M Sudheep Elayidom
Class Co-ordinator

Acknowledgment

I would like to convey my heartfelt gratitude to **Cognifyr.co**, an innovative and rapidly expanding service-based startup, for granting me the opportunity to intern and gain real-world exposure to professional software development. This internship has been an immensely enriching experience, enabling me to grow both technically and personally.

I am especially thankful to **Mr. Ajnas N B**, CEO of Cognifyr, for his constant support, guidance, and encouragement. His expertise, constructive feedback, and openness to ideas fostered a nurturing environment where learning was both challenging and enjoyable. I was given the chance to work on live projects, contribute meaningfully, and witness firsthand how technology is applied to create scalable, user-friendly solutions.

I am profoundly grateful to **Dr. Pramod Pavithran**, Head of Division, and **Dr. Sudheep Elayidom**, Class Coordinator, for their academic mentorship and support, which enabled me to pursue and successfully complete this internship as part of my curriculum at Cochin University of Science and Technology.

I also wish to extend sincere thanks to my parents, friends, and peers for their unwavering motivation and faith in my abilities. Their encouragement kept me focused and determined throughout the internship period.

Lastly, I would like to thank the entire team at Cognifyr for trusting me with significant responsibilities and making me feel like a valued contributor. This opportunity has undeniably played a crucial role in my development as an aspiring engineer and has deepened my passion for creating meaningful and impactful digital solutions.

Abstract

This report presents a comprehensive account of my one-month internship at Cognifyr.co, a service-oriented tech startup specializing in web and app development. Conducted from May 21 to June 30, 2025, the internship offered me an excellent platform to apply classroom learning to real-world projects, work with industry-standard tools and frameworks, and collaborate with professionals in a dynamic startup environment.

During this time, I contributed to two major projects for ONES, an ed-tech company. My responsibilities included revamping the backend of their admin dashboard using Python Django and Vue.js, resulting in significant improvements to its user interface and user experience. In addition, I developed essential APIs and integrations for their student-facing website using Postman and PgAdmin. For both projects, I also assisted in integrating frontend components with backend services to ensure smooth and reliable functionality.

The internship gave me valuable hands-on experience in backend development, system design, collaborative teamwork, API management, and version control tools. Beyond strengthening my technical expertise, it also enhanced my professional discipline, time management skills, and problem-solving capabilities. Overall, this internship has been an important milestone in my path toward becoming a proficient and confident software developer.

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1.Introduction

Internships act as a crucial bridge between academic learning and practical industry exposure. They provide students with opportunities to apply theoretical concepts in real environments and to gain a clearer understanding of how technology addresses real-world problems. As a fifth-semester Computer Science and Engineering student at Cochin University of Science and Technology, I had the privilege to intern at Cognifyr.co, a dynamic and fast-growing service-based startup.

The internship, from May 21 to June 30, 2025, immersed me in practical software development. I worked on real-time industry projects that sharpened my technical proficiency and taught me the importance of teamwork, version control, client-focused development, and intuitive user design.

During this period, I contributed significantly to two major projects. These projects were critical to the client's operations and required quality, precision, and performance. The two main projects were for ONES, an ed-tech company — I overhauled their broken admin dashboard using Vue.js and Python Django. And also integrated their student website with APIs and PostgreSQL using Postman and PgAdmin. I also assisted in frontend-backend integration for both projects to ensure smooth functionality.

This report offers a comprehensive overview of my time at Cognifyr.co, outlining the projects I worked on, the technologies I explored, the challenges I encountered, and the lessons I learned. The internship has built a strong foundation for my journey into software engineering by reinforcing my technical knowledge and boosting my confidence in managing real-world development tasks and growth.

2. Company Overview

Cognifyr.co is a rising service-based technology startup specializing in web development, mobile app development, UI/UX design, and digital automation solutions. Guided by a mission to ignite innovation through technology, Cognifyr is dedicated to delivering high-quality digital products customized to client needs, while upholding a strong commitment to affordability and excellence.

The startup distinguishes itself by offering aspiring developers and designers the chance to work on live client projects from the very first day. Known for its flexible work culture, strong mentorship, and open communication channels, Cognifyr provides an ideal environment for tech enthusiasts to learn and grow. During my internship, I was welcomed into a collaborative workspace where my contributions directly impacted core projects. The company's culture fostered continuous learning, accountability, and experimentation.

A key client I worked with during my internship was ONES, an ed-tech platform dedicated to transforming education through technology. Cognifyr's projects for ONES included overhauling their admin dashboard and creating a new student-facing portal, both requiring in-depth frontend development, responsive design, and API integration. This real-world engagement with client expectations and agile development workflows deepened my understanding of how startups function, adapt to evolving requirements, and strike the right balance between innovation and usability.

Cognifyr's dedication to guiding interns and involving them in impactful, client-focused projects makes it a truly rewarding place for aspiring engineers. The company values not only technical expertise but also initiative, creativity, and the willingness to take ownership—helping interns grow into capable and confident professionals.

3. Objectives of the Internship

The primary objective of my internship at Cognifyr.co was to gain practical, in-depth experience in backend development, API creation and management, database handling, and seamless frontend–backend integration. This internship aimed to bridge the gap between my academic knowledge and its application in real-world software development environments, with a particular focus on building scalable, secure, and maintainable systems.

A major goal was to strengthen my understanding of backend technologies, specifically server-side development using Python Django. I sought to deepen my knowledge of structuring robust backend logic, managing application workflows, implementing authentication and authorization mechanisms, and optimizing performance for production environments.

Another key objective was to enhance my skills in API development and integration. This involved learning to design, develop, test, and document RESTful APIs that enable smooth communication between client applications and server-side services. Special attention was given to ensuring APIs followed best practices for scalability, security, and maintainability, using tools such as Postman for testing and validation.

Database management formed a crucial part of my learning goals. I aimed to gain hands-on experience in designing, querying, and maintaining relational databases using PostgreSQL and PgAdmin. This included working with database schemas, writing optimized queries, ensuring data integrity, and integrating database operations with backend logic to support real-time application functionality.

Finally, a significant objective was mastering frontend–backend integration to ensure seamless data exchange and system coherence. This required close coordination between frontend and backend components, enabling efficient data flow, error handling, and synchronization across different layers of the application.

By focusing on these areas, the internship intended to equip me with practical skills that extend beyond theoretical coursework, preparing me to contribute effectively to production-grade software systems in professional settings.

4. Projects

During my internship at Cognifyr.co, I worked on two major client projects and one minor automation task, which greatly enhanced my skills as a web developer. The main projects were for ONES, an educational technology startup, while the smaller task involved creating an internal tool to streamline job opportunity discovery. My responsibilities focused on backend development, database management, API integration, and real-time data handling. Using Python Django, PostgreSQL, and Postman, I implemented server-side functionality, managed database operations, and integrated APIs to ensure smooth interaction between application layers. This work improved my ability to debug complex systems, optimize performance, and deliver production-ready features under strict deadlines. I also gained valuable experience in agile workflows, adapting quickly to changing requirements, and collaborating effectively with team members. Overall, the internship strengthened my technical expertise and prepared me to contribute confidently to future professional software projects.

4.1. ONES Admin Dashboard

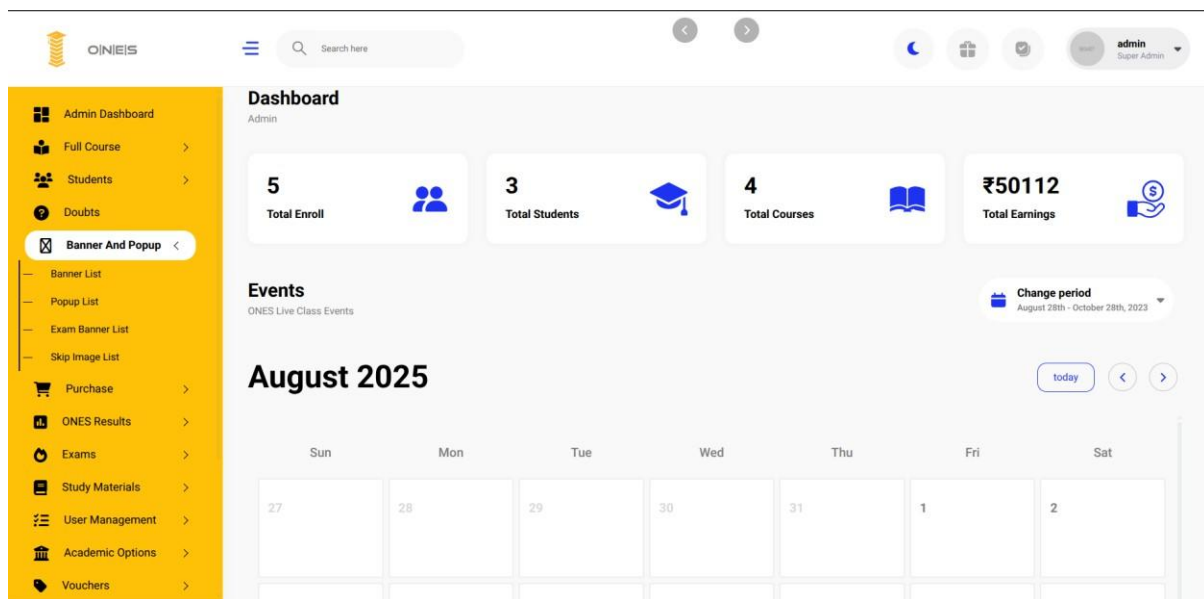


Fig 4.1 Ones Admin Dashboard

Technology Stack : Python Django, HTML, CSS, Vue.js, Git, REST API, PostgreSQL

4.1.1 Overview:

The ONES admin dashboard previously suffered from inefficient backend logic, frequent bugs, and poor data handling, which hindered the client's internal team from managing students and content effectively. My responsibility was to overhaul the backend architecture, optimize database queries, improve API performance, and ensure seamless integration with the frontend for reliable, scalable, and efficient operations.

4.1.2 Contributions:

- Refactored and optimized backend logic to support critical features such as user management, data updates, and analytics retrieval.
- Rebuilt and standardized API endpoints to ensure consistent, secure, and efficient communication between frontend and backend.
- Enhanced database queries and schema relationships to improve performance and reduce load times.
- Single-handedly debugged and resolved more than 60 critical bugs and performance issues, ensuring stable and reliable system operations.
- Integrated backend APIs with the frontend, enabling smooth execution of key functionalities and accurate real-time data display.
- Performed extensive manual testing and collaborated closely with the frontend team to ensure seamless data flow and synchronization across all components.

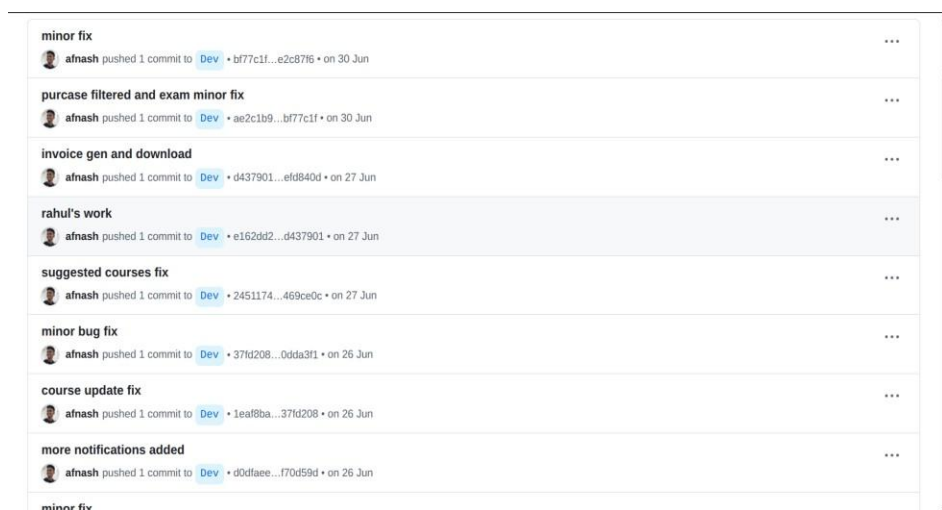


Fig 4.1.2 Github activity

4.2 ONES Student Website

ONES required a modern, student-facing website with reliable delivery of learning materials and account services

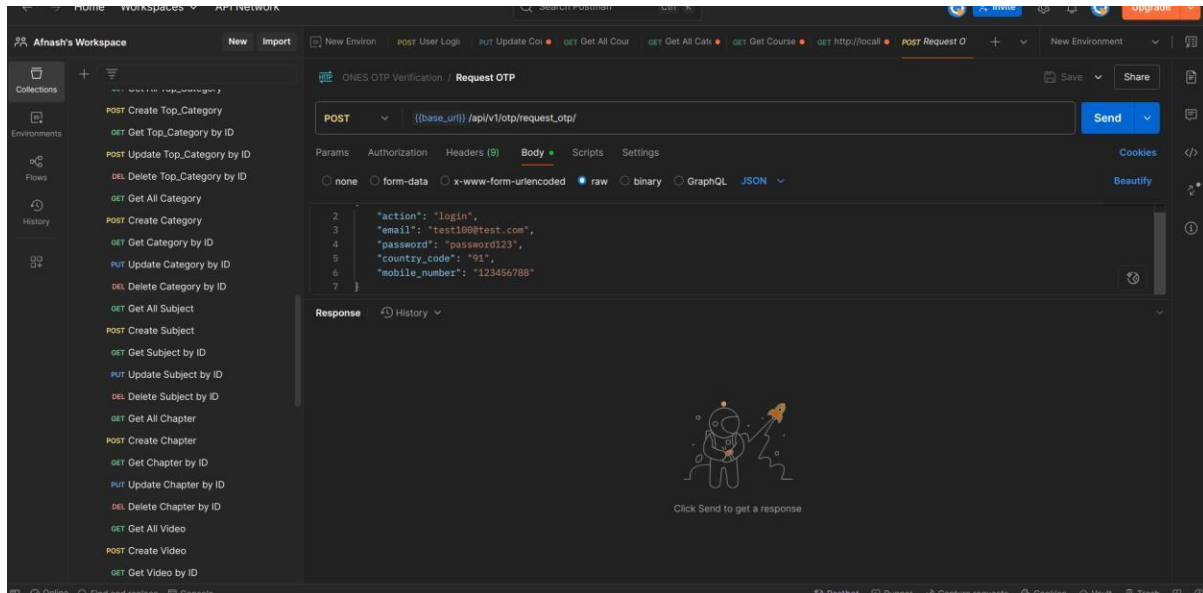


Fig 4.2 (a) API creation and testing in Postman

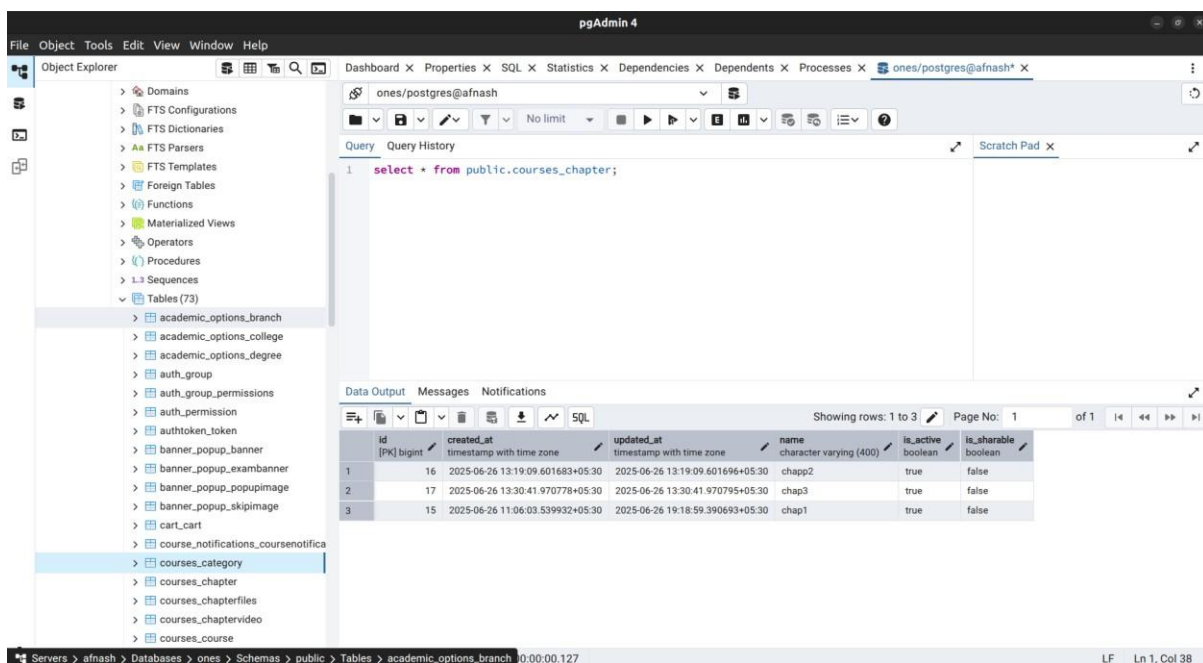


Fig 4.2 (b) Database management in PgAdmin

Technology stack : REST API, Postman, Github, PostgreSQL, Docker

4.2.1 Overview:

ONES required a modern, student-facing website with reliable delivery of learning materials and account services. I developed the backend using Docker for containerization, PgAdmin for database management, and Postman for API testing, ensuring scalable architecture, secure endpoints, and smooth frontend integration.

4.2.2 Contributions:

- Designed and implemented backend architecture to support dynamic routing and secure user authentication.
- Built and optimized RESTful API endpoints for student data retrieval, course management, and account services.
- Utilized Docker for containerized development and deployment, ensuring consistency across environments.
- Managed and optimized PostgreSQL databases via PgAdmin, including schema creation, indexing, and query tuning.
- Tested and validated API functionality with Postman to ensure accuracy, performance, and security.
- Collaborated with the frontend team to integrate APIs seamlessly, enabling real-time and reliable data flow.

Working on these two major projects gave me extensive hands-on experience in backend development, API integrations, and database management. I refactored legacy backend systems, implemented secure authentication flows, and optimized **PostgreSQL** databases using **PgAdmin** for improved performance and reliability.

Both projects involved close frontend-backend integration, where I built and tested **RESTful APIs** using **Postman** to ensure seamless data exchange. I also containerized services with **Docker**, enabling consistent development and deployment across environments. These experiences deepened my understanding of backend architecture, modern tooling, and delivering production-ready solutions under real client requirements.

5. Tools and Technologies used

During my internship at **Cognifyr.co**, I worked with a diverse range of industry-standard tools and technologies that spanned development, testing, version control, deployment, and team collaboration. This experience gave me hands-on exposure to the complete modern web development workflow, helping me build both technical proficiency and practical project execution skills.

5.1 Frameworks and Libraries

- **Python Django**

Django powered the backend of the ONES admin dashboard, where I fixed API issues, optimized database queries, and ensured smooth frontend-backend communication. I debugged authentication flows, resolved serialization errors, and updated views and models for accurate data handling.



Fig 5.1.1 Django logo

- **Vue.js:**

Vue was used in the ONES admin dashboard, where my role focused on identifying and fixing bugs within the existing codebase. I worked selectively on specific components, applying targeted fixes to resolve functionality issues and improve data flow. My contributions involved debugging Vue's component structure, adjusting state handling, and ensuring smoother interactions without major structural rewrites.



Fig 5.1.2 Vue.js logo

5.2 Programming Languages

- **Python:**

Used in Django backend for implementing server-side logic, APIs, authentication, and database operations. Enabled efficient debugging and rapid feature updates.



Fig 5.2.1 Python logo

- **HTML and CSS**

HTML was used to structure backend-rendered templates, while CSS ensured visual consistency and responsiveness across different screen sizes. I integrated semantic HTML elements for better accessibility and maintainability, and applied modular CSS to reduce redundancy



Fig 5.2.2 HTML and CSS logo

- **Javascript**

In the Vue.js environment, JavaScript was used to handle component logic, manage state, process API responses, and implement event handling, enabling smooth real-time data updates and seamless backend integration. Its role alongside HTML and CSS was crucial, as HTML provided the structural foundation, CSS ensured consistent and responsive styling, and JavaScript added the interactivity and dynamic behavior that brought the interface to life.

5.3 Version control and Containerization

- **Git and Github**

Git provided distributed version control for safe branching, commits, and history tracking, while GitHub served as the remote collaboration platform—hosting repositories, pull requests, code reviews, and issue tracking—enabling coordinated development, continuous integration, and reliable deployment workflows.

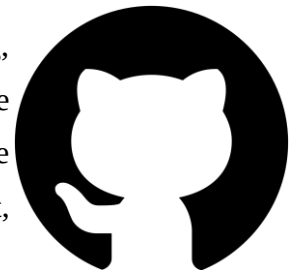


Fig 5.3.1 Github logo

- **Docker**

Docker was used to containerize the application, ensuring consistent development and deployment environments. It simplified dependency management, isolated services, and streamlined testing and deployment across different systems without configuration conflicts.

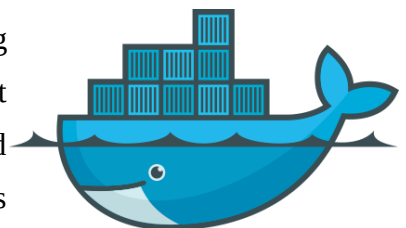


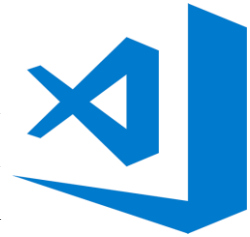
Fig 5.3.2 Docker logo

5.4 Development Tools

- **VS code**

Served as the primary code editor, offering syntax highlighting, debugging tools, and extensions that streamlined backend development and API integration tasks. Its integrated terminal and Git support made coding, testing, and version control seamless in a single environment.

Fig 5.4.1 VS Code logo



- **Postman**

Used for testing and validating RESTful APIs, enabling efficient debugging, request automation, and response verification during backend–frontend integration. It also supported environment variables and collections, making API testing more organized and reusable.



Fig 5.4.2 Postman logo

- **PgAdmin**

Provided a graphical interface for managing PostgreSQL databases, allowing efficient querying, schema management, and data inspection for backend development. Its visualization and management tools made handling complex datasets and relationships more intuitive.



Fig 5.4.3 pgAdmin logo

This stack of languages, frameworks, and tools created a cohesive workflow, enabling smooth execution from development to deployment. Python, JavaScript, HTML, and CSS handled core functionality, while Django and Vue.js facilitated reliable backend–frontend connectivity. Git and GitHub supported collaborative version control, Docker provided uniform deployment environments, and utilities like Postman, pgAdmin, and VS Code enhanced API validation, database handling, and coding efficiency. Combined, they ensured the projects were robust, maintainable, and adaptable to evolving requirements.

6. Key Learnings

My internship at Cognifyr.co marked a pivotal phase in my professional development, giving me hands-on exposure to live systems, demanding timelines, and complex technical challenges.

6.1. Technical Expertise

- **Backend Connectivity & APIs:** Deepened my knowledge of RESTful API design, data exchange, and secure authentication. Utilized Axios, Postman, and pgAdmin to streamline integration, validate endpoints, and monitor data flow. This allowed me to deliver reliable, production-ready integrations that scaled well under varying loads.
- **Database Operations:** Worked with PostgreSQL to structure, query, and optimize data, ensuring seamless interaction between the backend and client-facing features. Also implemented indexing and query optimization to improve response times for high-traffic endpoints.
- **Containerization & Deployment:** Applied Docker to create consistent, reproducible environments for both development and production stages. This minimized environment-specific bugs and simplified deployment pipelines.
- **Legacy Code Maintenance:** Analyzed existing Django and Vue.js modules, resolving over 60 functional and integration issues to enhance stability.

6.2 Workflow & Delivery

Balanced overlapping tasks across sprints by setting clear priorities, breaking work into achievable goals, and maintaining delivery precision. Regular progress tracking ensured that deadlines were consistently met without sacrificing quality.

6.3 Team Synergy

Collaborated with cross-functional teams to refine requirements, troubleshoot blockers, and ensure deliverables met client expectations. Frequent check-ins promoted alignment and reduced miscommunication in fast-moving projects.

6.4 Debugging and Issue Resolution

Addressed production-level bugs using targeted diagnostics with browser DevTools and framework-specific debugging tools, ensuring minimal downtime. Each fix was documented for future reference, reducing repeated troubleshooting efforts.

7. Challenges Faced

7.1. Navigating and Improving Legacy Backend Code

Working with a cluttered Django backend in the ONES admin dashboard was challenging due to messy code and poor documentation. This pushed me to:

- Trace complex models, views, and serializers to understand data flow and dependencies.
- Apply incremental refactoring to improve code quality without breaking existing features.

7.2. Managing APIs with Postman and PostgreSQL

Building and testing APIs involved:

- Writing efficient queries and managing migrations in PostgreSQL using pgAdmin.
- Ensuring API endpoints handled CRUD operations with proper validation and error handling.

7.3. Containerizing and Deploying with Docker

Docker helped streamline development and deployment by:

- Containerizing the Django backend and PostgreSQL database for reproducible environments.
- Writing docker-compose configurations for multi-container orchestration.

7.4. Balancing Speed and Stability in a Startup Backend

The fast-paced startup environment meant I had to:

- Rapidly implement backend features with solid, scalable code architecture.
- Prioritize bug fixes and maintain API stability under tight deadlines.
- Continuously test APIs and database interactions to avoid regressions while iterating quickly.

8. Conclusion

My internship at Cognifyr.co was a pivotal and highly enriching experience that deepened my backend development skills and exposed me to real-world challenges. Working extensively with Django, PostgreSQL, Docker, and API testing tools like Postman, I gained hands-on experience in building robust, scalable, and maintainable backend systems.

From debugging legacy Django codebases and refactoring complex models and serializers to managing PostgreSQL databases through pgAdmin, I learned the importance of clean architecture and incremental improvements. Containerizing applications with Docker not only streamlined development but also prepared me for smooth deployment workflows.

A major highlight was deploying backend services on AWS, where I navigated cloud infrastructure challenges, automated deployments, and optimized performance in production environments. This end-to-end exposure—from local development to live cloud deployment—gave me a comprehensive understanding of backend lifecycle management.

Beyond technical skills, this internship reinforced the value of systematic debugging, thorough API testing, and maintaining stable releases even under tight deadlines. Collaborating closely with frontend teams, designers, and mentors gave me insight into effective communication and agile workflows in a professional setting.

I leave this internship with:

- A solid command of backend frameworks and tools such as Django, PostgreSQL, Docker, and AWS,
- Improved confidence in debugging, deploying, and maintaining complex systems,
- A stronger appreciation for cross-team collaboration and continuous feedback, and
- A clear career direction focused on backend development and cloud technologies.

I'm deeply grateful to the Cognifyr team for their support, challenges, and trust, which fostered an environment of continuous learning and meaningful contribution. This experience has shaped my mindset around responsibility, quality, and growth—principles I will carry forward in my tech career.

References

1. **Cognifyr**: <https://cognifyr.co> – An emerging service-based startup focused on delivering innovative web, app, and design solutions tailored to client needs.
2. **ONES Coaching**: <https://www.onescoaching.com> – An educational platform that provides coaching, mock tests, and learning tools for students and competitive exam aspirants.
3. **Docker**: <https://www.docker.com> – A leading containerisation platform for developing, shipping, and running applications consistently across environments.
4. **PostgreSQL**: <https://www.postgresql.org> – An advanced, enterprise-class open-source relational database system known for stability and extensibility.
5. **EmailJS**: <https://www.emailjs.com> – A client-side email service that enables sending emails through JavaScript without a server backend.
6. **Next.js**: <https://nextjs.org> – A React-based framework for building high-performance web applications with server-side rendering and static site generation.
7. **Vue.js**: <https://vuejs.org> – A progressive JavaScript framework for building user interfaces with a flexible and component-based architecture.
8. **GitHub**: <https://github.com> – A platform for version control and collaboration, allowing developers to manage code, track issues, and contribute in teams.
9. **Postman**: <https://www.postman.com> – A collaboration platform for API development that simplifies testing, debugging, and documenting RESTful services.
10. **ChatGPT** (OpenAI): <https://chat.openai.com> – An AI-powered assistant developed by OpenAI used for programming help, problem-solving, and productivity enhancement.
11. **Gemini** (Google AI): <https://gemini.google.com> – Google’s conversational AI and research tool offering language-based capabilities for writing, coding, and brainstorming ideas.